Synchronous phasors and the power angle

Practice questions · Day 14 · Fri 6 Nov · pack M11

5. Practice

From your course files

PQ 8.8 (practice questions). A round-rotor synchronous generator has a synchronous reactance of 1.8 per unit. Its excitation system is controlled so that its terminal voltage is always 1 per unit. Resistance and saturation may be neglected.

(a) What excitation (in per unit) is required to produce the generator's rated volt-amp output at unity power factor?
(b) What is the rotor angle at this loading?
(c) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power that can be delivered without losing synchronism?

Jun24 Q4b (past exam, 6 marks). *4b) Synchronous Machines [6 marks]* A round-rotor synchronous generator has a synchronous reactance of 1.8 per-unit. Its excitation system is controlled so that its terminal voltage is always 1 per-unit. Resistance and saturation may be neglected.

i) What excitation (in per unit) is required to produce the generator's rated volt-amp output at 0.9 leading power-factor? [3 marks]

ii) What is the rotor angle at this loading? [1 mark]

iii) If both the terminal voltage and the excitation are fixed at the above levels, what is the theoretical maximum power, in per-unit, that can be delivered without losing synchronism? What is the rotor angle corresponding to this condition? [2 marks]

PQ 8.9 (practice questions). A 200 MVA generator has an unsaturated synchronous reactance of 1.8 per unit. If the excitation is set at a level which would give an open-circuit voltage of 1.5 per unit, and saturation and resistance can be neglected, what is the power in

megawatts that can be delivered when the machine's terminal voltage is 2% above its rated value and the rotor angle is not to exceed 70°?

PQ 8.11 (practice questions). A power station generator is rated at 250 MVA, 15 kV and has a star-connected stator winding. After manufacture, an open-circuit test is performed and a line voltage of 5.75 kV is measured with an excitation of 50 A. Under the same value of excitation, a current of 2050 A is measured with the short-circuit test. The resistance of each stator phase is 0.15 ohm. The generator is supplying its full-load rating at nominal voltage and a power factor of 0.85 lag. Ignoring the effects of magnetic saturation, calculate :

(a) The unsaturated synchronous reactance in per-unit
(b) The per-unit stator current magnitude and phase relative to the terminal voltage
(c) The induced voltage E_0 in per-unit showing both its magnitude and phase angle [Hint : draw the synchronous machine equivalent circuit and hence see that $E_0 = V + I(R + jX_S)$]
(d) The corresponding value of E_0 as a line voltage. [Hint : multiply the magnitude of the per-unit value by the base line voltage]
(e) The excitation required in amps.
(f) The rotor angle?

PQ 8.12 (practice questions). A 120 MVA, 50Hz, 3,000rpm, 16 kV power station generator gives the following test results :

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
50	4.4	600

The stator resistance is measured as 0.1 ohm. Magnetic saturation can be ignored.

(a) From the test results, calculate the values of the unsaturated synchronous reactance in per unit
(b) Calculate the stator current magnitude and phase in per-unit when the generator is supplying a load of 102 MW at a power factor 0.95 lag when the terminal voltage is at its rated

value. Use the terminal voltage as the reference phasor.
(c) Calculate the required excitation (in amps) for the situation in part (b). What is the ratio of this value of excitation current compared to the excitation current required for synchronization?

PQ 8.13 (practice questions). The stator winding of a synchronous motor is rated at 330 kVA. If the motor is supplying a mechanical load of 270 kW at an efficiency of 92%, what is the maximum reactive power it can provide for power factor correction?

PQ 8.14 (practice questions). A production line uses an induction motor to drive a 200 kW mechanical load. The induction machine has an efficiency of 90% and a power factor of 0.82. A 100 kVA synchronous motor is used to drive an 80 kW load nearby. Assume the synchronous motor has an efficiency of 100%. What is the highest input power factor of the two machines combined as a system?

Jun23 Q4b (past exam, 7 marks). *4b) Power Factor Correction [7 marks]* An industrial plant uses two identical induction motors, each driving a mechanical load of 200kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i) Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii) Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]

iii) Use the above results to determine the maximum power factor for the plant. [3 marks]

PQ 8.15 (practice questions). An industrial installation has an existing load of 2760 kVA at a power factor of 0.71 lag. A major extension is planned, requiring 1500 kW of mechanical load. It is decided to satisfy 800 kW of this demand using induction motors, and to purchase synchronous motors to supply the remainder. The parameters of the additional motors are as follows :

Induction Motors : Efficiency 89% : Power factor 0.75 lag
Synchronous Motors : Efficiency 93% : Total rating 1350 kVA

What is the maximum predicted power factor of the completed installation?

PQ 8.10 (practice questions). A synchronous generator has a rating of 150 MVA and a rated terminal voltage of 15 kV. It's stator is star-connected. It is given the standard tests after manufacture with the following results :

Open-Circuit Test : Slope of linear portion = 500 volts (line) per field amp.
Short-Circuit Test : Slope = 100 amps per field amp

The stator resistance is measured as 0.3 Ω / phase. Magnetic saturation can be neglected.

(a) Find the base current and base impedance. [Hint : choose the machine rated VA and voltage as the base power and base voltage.]
(b) What is the value of the unsaturated synchronous reactance *in per-unit* ? [Hint : choose any value of field current, say 1A and calculate the open-circuit voltage and short-circuit current corresponding to this. Find the synchronous impedance and hence the synchronous reactance from the stator resistance.]
(c) What value of field current would be required for synchronization? (Note that to synchronize the machine to the system, the open-circuit voltage E_0 is set to rated voltage.)
(d) At synchronization, what is the value of the open-circuit voltage E_0 in per-unit?
(e) Under a certain operating condition, the open-circuit voltage E_0 is calculated to be 1.65 pu. What field current (in amps) would be required to produce this?

Jun23 Q4c (past exam, 8 marks). *4c) Short-Answer Questions [8 marks]* Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) Explain the concept of power-factor correction. Why is power-factor correction useful in power systems? [3 marks]

ii) Carefully explain the differences and similarities between transformers and induction machines. [3 marks]

iii) Consider a squirrel-cage induction machine. If the rotor cage was made from a perfect conductor (with infinite conductivity) how would this affect the properties of the machine? [2 marks]

Jun24 Q4c (past exam, 9 marks). *4c) Short-Answer Questions [9 marks]* Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) We have discussed three types of electric power: real, reactive and apparent. Explain briefly what each one is and their practical significance. [3 marks]

ii) Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power. [3 marks]

iii) Carefully explain why the input current of an induction machine under no-load conditions is not zero. What are the physical effects which cause this? [3 marks]

Practice Questions, Ch8 short answers (j) and (k) (bonus). From “Some Short-Answer Questions” at the end of Chapter 8:

j) With the aid of sketch, explain what limits the reactive power capability of a synchronous machine.

k) Why is the power-factor of industrial plants usually lagging? Why are synchronous motors sometimes used in such plants?

New questions

N1. [3 marks] A 50 MVA synchronous compensator (P = 0) has $X_S = 1.5$ pu and is connected to a bus held at 1 pu. Neglect R.

a. What E_0 is needed for it to supply 0.8 pu reactive power to the bus? Is it over- or under-excited? [2]
b. What E_0 is needed for it to absorb 0.5 pu? [1]

N1 circuit: compensator E0 behind j1.5 pu on a 1 pu bus

N2. [4 marks] A 50 MVA, 11 kV star-connected generator has $X_S = 2.4\ \Omega$ per phase. Its excitation gives $E_0 = 14$ kV (line). It runs at $\delta = 25^\circ$ on an 11 kV bus. Neglect R.

a. Find the three-phase output power in MW using per-phase quantities, then check it with line quantities. [2]
b. Find the maximum power at this excitation, in MW and in per-unit. [2]

N2 circuit: E0 = 14 kV line behind j2.4 ohm on an 11 kV bus

N3. [6 marks] A generator with $X_S = 1.6$ pu delivers S = 0.8 pu at V = 1 pu. Neglect R.

a. Find E_0 and δ at 0.85 pf lagging. [2]

b. The real power is held constant, and the excitation is changed until the pf is 0.85 leading. Find the new E_0 and δ. [2]
c. Find P_{max} in each case, and comment on stability. [2]

N3 circuit: E0 behind j1.6 pu, V = 1 pu, S = 0.8 pu

N4. [5 marks] A generator ($X_S = 1.8$ pu, V = 1 pu) supplies a constant P = 0.6 pu. The operator lowers the excitation to absorb reactive power. Neglect R.

a. What is the minimum E_0 if δ must not exceed 75°? [2]
b. At that point, find the stator current phasor, the reactive power, and the pf. [3]

N4 circuit: E0 behind j1.8 pu, V = 1 pu, P = 0.6 pu

N5. [10 marks] A plant takes 600 kW at 0.75 pf lagging from a 415 V supply. A new 400 kVA, 415 V synchronous motor ($X_S = 1.0$ pu on its own rating, η = 95%) is installed to drive a 300 kW load. It is run fully over-excited to correct the pf. Neglect R in the phasor model, and take the motor's electrical input as the phasor power.

a. Find the motor's input power and its maximum reactive power. [2]
b. Find the plant's new pf. [2]
c. On the motor base, find the motor current phasor (in pu and in amps) and the required excitation E_0 (in pu and as a line voltage). [4]
d. The excitation is then reduced until the motor runs at unity pf. Find the new E_0 and the plant pf. [2]

N5 plant: 600 kW at 0.75 lag in parallel with a 400 kVA synchronous motor

6. Quick recall check

1. At what rotor angle does a round-rotor machine deliver maximum power, and what happens if you try to go past it?
2. A generator's output current is leading. Is it over- or under-excited, and does it supply or absorb Q?
3. Does the per-unit $P = \frac{VE_0}{X_S}\sin\delta$ give per-phase or three-phase power?
4. A 100 kVA synchronous motor drives 60 kW at 100% efficiency. What is the most leading reactive power it can supply?
5. On a V-curve at constant P, where is the stator current a minimum, and what is the pf there?